

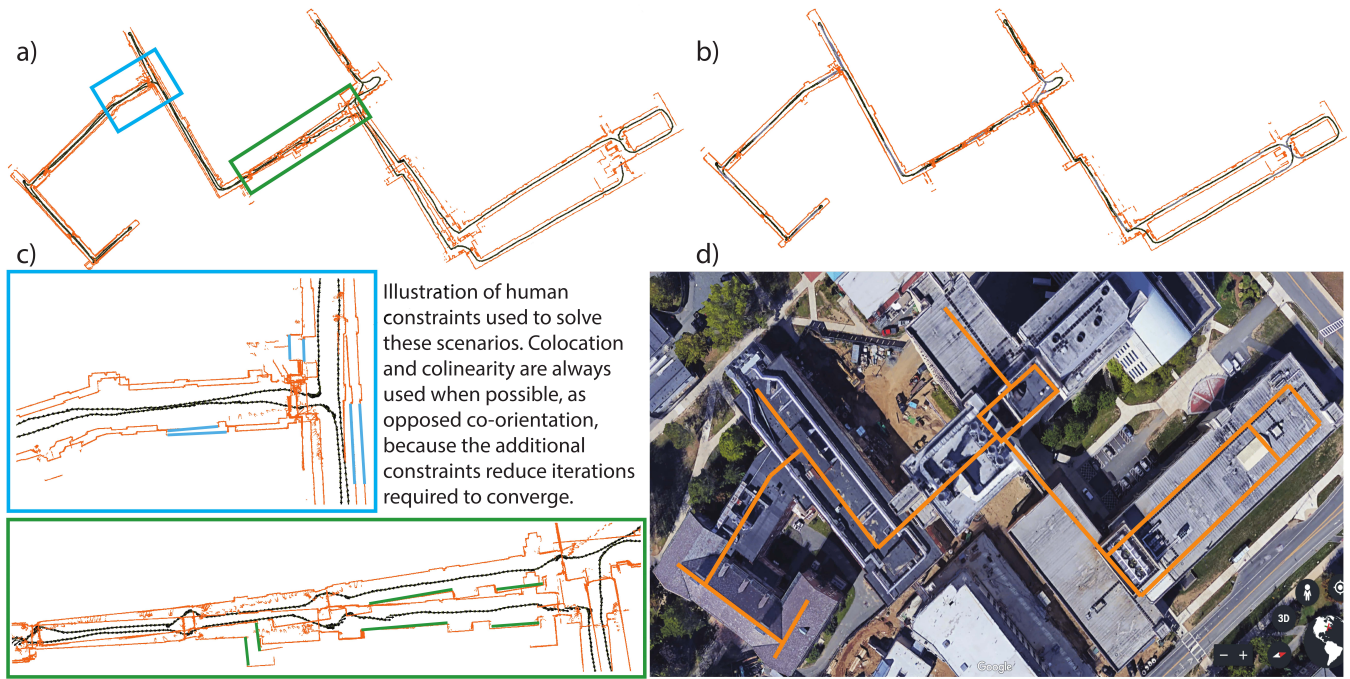


Figure 7: A large map a) corrected by HitL-SLAM b) using human correction, some of which are highlighted c). d) shows an approximate overlay of the map onto an aerial image of the complex from google earth. The map contains over 3000 poses.

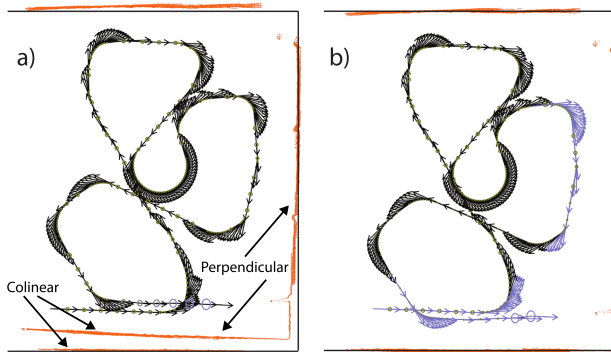


Figure 8: Initial a) and final b) maps for the 'lost poses' experiment. Observations are shown in orange, poses are black arrows, and ground truth (walls) is represented by the black lines. Poses involved in human constraints are colored blue.

framework for interpreting human input using the EM algorithm, as well as a factor graph based algorithm for incorporating human input into pose-graph SLAM. Future work in this area could proceed towards further reducing the human requirements, and extending this method for higher dimensional SLAM and for different sensor types.

### Acknowledgments

The authors would like to thank Manuela Veloso from Carnegie Mellon University for providing the CoBot4 robot used to collect the AMRL datasets, and to perform experiments at University of Massachusetts Amherst. This work

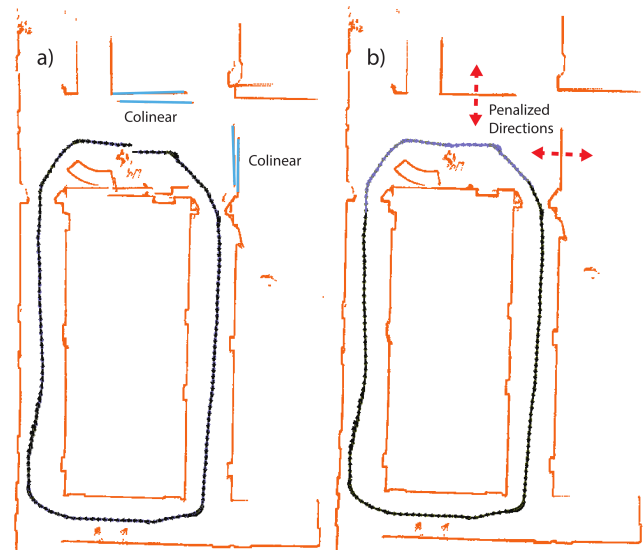


Figure 9: Example of how rank-deficient (collinear) constraints can be combined to effect a full rank constraint. Often, it is easier to tell if two sets of observations are collinear rather than colocated. Supporting rank-deficient constraints allows HitL-SLAM to operate on a larger set of maps.

is supported in part by AFRL and DARPA under agreement #FA8750-16-2-0042.

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